

# HDOC Day-14 Activity

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## Menu-Driven C Program: Unique Digits & Digit Frequency

### Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique.
- Find the frequency of each digit from 0 to 9.

### Algorithm

1. Start.
2. Input a positive integer num.
3. If  $\text{num} \leq 0$ , display "Please enter a positive integer" and stop.
4. Display the menu: 1. Check whether all digits are unique 2. Find frequency of each digit.
5. Input the user's choice.
6. Use switch(choice) to perform the selected operation.
7. For Case 1, initialize  $\text{freq}[10]$  to 0 and set  $\text{temp} = \text{num}$ . Extract each digit using  $\text{digit} = \text{temp} \% 10$ . If  $\text{freq}[\text{digit}]$  is already 1, mark the number as not unique; otherwise set  $\text{freq}[\text{digit}] = 1$ . Update  $\text{temp} = \text{temp} / 10$ . After all digits are processed, display "Unique" or "Not Unique".
8. For Case 2, initialize  $\text{freq}[10]$  to 0 and set  $\text{temp} = \text{num}$ . Extract each digit, increment  $\text{freq}[\text{digit}]$ , and update  $\text{temp} = \text{temp} / 10$ . Display the frequency of digits 0 through 9.
9. For any other choice, display "Invalid choice".
10. Stop.

### C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num, temp, digit, choice, i;
```

```
    int freq[10] = {0};
```

```
    int unique = 1;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &num);
```

```

if (num <= 0)
{
    printf("Please enter a positive integer.\n");
    return 0;
}

printf("\nMENU\n");
printf("1. Check whether all digits are unique\n");
printf("2. Find frequency of each digit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

temp = num;

switch (choice)
{
    case 1:
        while (temp > 0)
        {
            digit = temp % 10;
            if (freq[digit] == 1)
            {
                unique = 0;
                break;
            }
            freq[digit] = 1;
            temp = temp / 10;
        }

        if (unique == 1)
            printf("All digits are Unique\n");
        else
            printf("All digits are Not Unique\n");
        break;

    case 2:
        while (temp > 0)
        {
            digit = temp % 10;
            freq[digit]++;
            temp = temp / 10;
        }

        for (i = 0; i <= 9; i++)
            printf("freq_%d = %d\n", i, freq[i]);
        break;
}

```

```

    default:
        printf("Invalid choice.\n");
    }

    return 0;
}

```

## Test Case Table

| Test Case | Input Number | Choice | Expected Output                                            | Actual Output                   | Result |
|-----------|--------------|--------|------------------------------------------------------------|---------------------------------|--------|
| 1         | 12345        | 1      | Unique                                                     | Unique                          | Pass   |
| 2         | 12234        | 1      | Not Unique                                                 | Not Unique                      | Pass   |
| 3         | 112333       | 2      | freq_0=0, freq_1=2, freq_2=1, freq_3=3, others=0           | Same as expected                | Pass   |
| 4         | 98760        | 2      | freq_0=1, freq_6=1, freq_7=1, freq_8=1, freq_9=1, others=0 | Same as expected                | Pass   |
| 5         | -25          | 1      | Please enter a positive integer                            | Please enter a positive integer | Pass   |
| 6         | 123          | 3      | Invalid choice                                             | Invalid choice                  | Pass   |

## Sample Output

### Compilation and Execution

**Compilation:** gcc day14.c -o day14

```

(kali㉿kali)-[~/hdoc14.c]
$ gcc day14.c -o day14

```

**Execution - Choice 1: Unique digits**

```

(kali㉿kali)-[~/hdoc14.c]
$ ./day14
Enter a positive integer: 12345

MENU
1. Check whether all digits are unique
2. Find frequency of each digit
Enter your choice: 1

All digits are Unique

```

### Execution - Choice 2: Frequency for 12345

```
(kali㉿kali)-[~/hdoc14.c]
$ ./day14
Enter a positive integer: 12345

MENU
1. Check whether all digits are unique
2. Find frequency of each digit
Enter your choice: 2
freq_0 = 0
freq_1 = 1
freq_2 = 1
freq_3 = 1
freq_4 = 1
freq_5 = 1
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

### Execution - Choice 2: Frequency for 112333

```
(kali㉿kali)-[~/hdoc14.c]
$ ./day14
Enter a positive integer: 112333

MENU
1. Check whether all digits are unique
2. Find frequency of each digit
Enter your choice: 2
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

Example 1: Number = 12345, Choice = 1 → Unique

Example 2: Number = 12234, Choice = 1 → Not Unique

Example 3: Number = 112333, Choice = 2 → freq\_0=0, freq\_1=2, freq\_2=1, freq\_3=3, freq\_4=0, freq\_5=0,  
freq\_6=0, freq\_7=0, freq\_8=0, freq\_9=0